EASYVNA ANTENNA REPORT

whip-A

EasyVNA Web
by PiX

SIMULATED DATA

Date	2026-11-07 17:29	Report	whip-A-20261107-172956
Device	NanoVNA-H 4	Firmware	1.2.50 · EasyVNA by PiX (EasyVNA Web mock)
Profile	869 MHz profile	Calibration	applied · slot 5 (device profile), 779–959 MHz

Verdict per band

869 MHz band — PASS

863–870 MHz

Worst SWR	Resonance	Return loss
1.22	**869.53 MHz**	**20.2 dB**
@ 863.00 MHz	+0.5 MHz vs 869 (in band)	worst in band

Mismatch loss	SWR < 2 bandwidth	Q (estimated)
0.04 dB	**49.6 MHz**	**12**
worst in band		

869.4–869.65 MHz sub-band — PASS

869.4–869.65 MHz

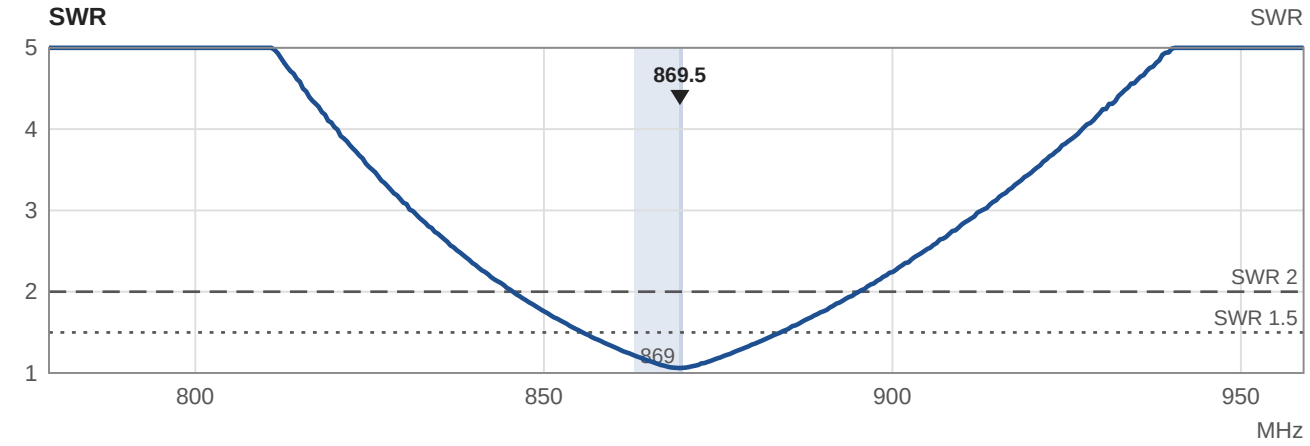
Worst SWR	Resonance	Return loss
1.06	**869.53 MHz**	**30.4 dB**
@ 869.65 MHz	0.0 MHz vs 870 (in band)	worst in band

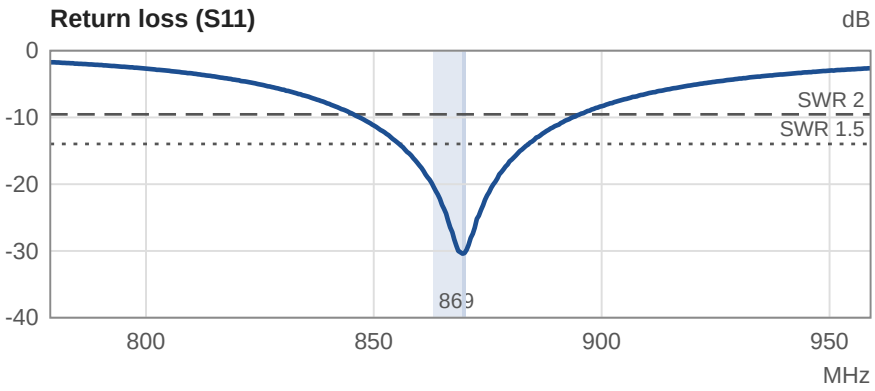
Mismatch loss
0.00 dB
worst in band

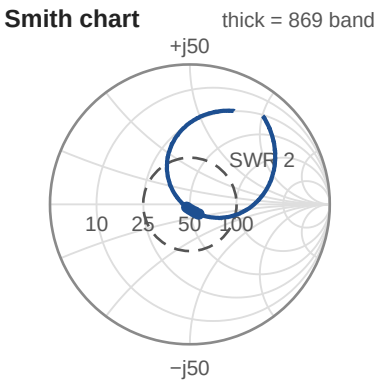
NOTES

868 MHz quarter-wave whip on a 50 × 50 mm ground plane, 10 cm RG316 pigtail. Simulated data.

869 MHz profile · 779–959 MHz · swept 17:29:55







Best SWR	1.06 @ 869.45 MHz	Return loss there	30.4 dB
Mismatch loss there	0.00 dB	Impedance there	48.2 − j2.3 Ω
SWR < 2 bandwidth	49.6 MHz	Q (estimated)	12
Sweep	779–959 MHz, 401 points	Calibration	calibrated for this range (slot 5 (device profile))

Resonances

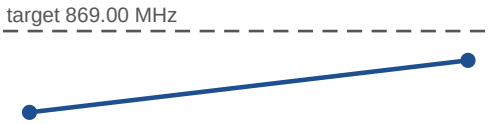
#	MHZ	SWR	RL	ML	Z	BW<2	Q	NOTE
1	**869.53**	1.06	30.4	0.00	48.2 − j2.3 Ω	49.6 MHz	12	In the 869 band (+0.5 MHz from 869.00 MHz).

HOLD: stable positions · 869 MHz profile · 779–959 MHz

Positions	**4**
Resonance spread	**6.3 MHz** (864.45–870.79 MHz)
Worst in-band SWR	**1.27 (position #4)**

#	CAPTURED	RESONANCE	BEST SWR	WORST IN BAND
1	17:29:21	866.10 MHz	1.08	1.15
2	17:29:24	868.96 MHz	1.08	1.21
3	17:29:30	864.45 MHz	1.08	1.19
4	17:29:36	870.79 MHz	1.08	1.27

Trim log · whip-A · target 869.000 MHz



#	TIME	LENGTH	CUT	RESONANCE	Δ TARGET	SWR @ TARGET
1	17:29:03	82.0 mm	–	856.644 MHz	-12.4 MHz	1.55
2	17:29:04	81.1 mm	0.9 mm	864.531 MHz	-4.5 MHz	1.33

Next: **cut 0.5 mm** → 80.6 mm

9.05 MHz/mm (fit, learnt from 2 logs)

S21 compare · relative to stock whip · 869 863.00–870.00 MHz

Fixed reference antenna on port 1, antenna under test on port 2. Relative numbers only (no THRU calibration needed); same placement and cables for every antenna.

#	ANTENNA	Δ IN BAND (AVG)	WORST	BEST	MEASURED
1	**tuned whip**	**+2.6 dB**	+2.6 dB	+2.7 dB	17:29:11
2	**stock whip**	0.0 dB	0.0 dB	0.0 dB	17:29:09
3	**helical 433**	**-0.6 dB**	-0.9 dB	-0.3 dB	17:29:14
4	**fake (dummy)**	**-37.4 dB**	-40.3 dB	-35.7 dB	17:29:16

Grades: PASS = worst in-band SWR < 1.5, OK < 2, FAIL otherwise (band edges interpolated). Q is estimated from the SWR < 2 bandwidth assuming a match at resonance. Mismatch loss = −10·log10(1−|Γ|²).